Supplementary Information for

TM-score-Comp: a quick and accurate method for assessing qualities of complex structure predictions of proteins, nucleic acids, and small molecule ligands

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Supporting Texts

Text S1. Distance-scale definitions d_0 for different molecule types

To compute the uniform TM-score in TM-score-Comp for complexes containing multiple molecular components, we employ molecule-type–specific normalization factors d_0, including $d_0^{protein}$, d_0^{DNA} and d_0^{RNA} and d_0^{ligand}. These normalization factors are designed to eliminate the length dependence of TM-score values for randomly related structure pairs across different molecule types[1-3]. The same functional form is used for DNA and RNA molecules.

The distance-scale definitions are given as follows:

$$\begin{cases} d_0^{protein} = \begin{cases} 1.24\sqrt[3]{L_{protein} - 15} - 1.8, & \text{if } L_{protein} > 21 \\ 0.5, & \text{otherwise} \end{cases} \\ d_0^{DNA} = \begin{cases} 0.6\sqrt{L_{DNA} - 0.5} - 2.5, & \text{if } L_{DNA} \geq 30 \\ 0.7, & \text{if } 24 \leq L_{DNA} \leq 29 \\ 0.6, & \text{if } 20 \leq L_{DNA} \leq 23 \\ 0.5, & \text{if } 16 \leq L_{DNA} \leq 19 \\ 0.4, & \text{if } 12 \leq L_{DNA} \leq 15 \\ 0.3, & \text{otherwise} \end{cases} \\ d_0^{RNA} = \begin{cases} 0.6\sqrt{L_{RNA} - 0.5} - 2.5, & \text{if } L_{RNA} \geq 30 \\ 0.7, & \text{if } 24 \leq L_{RNA} \leq 29 \\ 0.6, & \text{if } 20 \leq L_{RNA} \leq 23 \\ 0.5, & \text{if } 16 \leq L_{RNA} \leq 19 \\ 0.4, & \text{if } 12 \leq L_{RNA} \leq 15 \\ 0.3, & \text{otherwise} \end{cases} \\ d_0^{ligand} = \begin{cases} 0.55\sqrt[3]{L_{ligand} - 9} + 0.15, & L_{ligand} \geq 9 \\ 0.1, & otherwise \end{cases} \end{cases} \quad \text{(S1)}$$

where $L_{protein}$, L_{DNA}, L_{RNA} and L_{ligand} are the total lengths of proteins, DNAs, RNAs and small molecule ligands, respectively, in the target complex structure.

Text S2. Ligand RMSD

Ligand RMSD (LRMSD) is computed as the backbone RMSD of the ligand after superimposing the receptor structure. The receptor is defined as the chain with the greater number of amino acid/nucleotide residues; if both chains contain the same number of residues, the chain that making the lower LRMSD is selected as the receptor. In receptor-small molecule ligand complexes, LRMSD is evaluated over all ligand heavy atoms following superposition onto the receptor interface.

Text S3. Fraction of native contacts

The fraction of native contacts (F_{nat}) is defined as the ratio of the number of correctly predicted native contacts to the total number of contacts observed in the native interface[4]. An interface is defined as the set of interacting residues or nucleotides from two different molecules, where at least one pair of heavy atoms is within 5Å of each other. A predicted contact is considered correct if the corresponding pair of interfacial residues in the predicted complex also forms a contact in the native structure.

Text S4. Evaluation metrics for classification performance

To assess the performance of TM-score-Comp in classifying positive and negative prediction models, we employed six standard evaluation metrics commonly used in binary classification tasks: sensitivity (Sen), specificity (Spe), accuracy (Acc), precision (Pre), F1-score (F1), and Matthews' correlation coefficient (MCC). These metrics are defined as follows:

$$\begin{cases} \text{Sen} = \dfrac{\text{TP}}{\text{TP} + \text{FN}} \\ \text{Spe} = \dfrac{\text{TN}}{\text{TN} + \text{FP}} \\ \text{Acc} = \dfrac{\text{TP} + \text{TN}}{\text{TP} + \text{FN} + \text{TN} + \text{FP}} \\ \text{Pre} = \dfrac{\text{TP}}{\text{TP} + \text{FP}} \\ \text{F1} = 2 \cdot \dfrac{\text{Sen} \cdot \text{Pre}}{\text{Sen} + \text{Pre}} \\ \text{MCC} = \dfrac{\text{TP} \cdot \text{TN} - \text{FN} \cdot \text{FP}}{\sqrt{(\text{TP} + \text{FN}) \cdot (\text{TP} + \text{FP}) \cdot (\text{TN} + \text{FN}) \cdot (\text{TN} + \text{FP})}} \end{cases} \tag{S2}$$

Here, TP, FN, TN, and FP denote the numbers of true positives, false negatives, true negatives, and false positives, respectively. In this study, we employ the threshold corresponding the maximum MCC value to judge each sample is predicted as positive or negative one.

In addition to these threshold-dependent metrics, two threshold-independent summary measures—AUC (area under the receiver operating characteristic curve) and AUCPR (area under the precision–recall curve)—were also used to evaluate the overall classification performance.

Text S5. Interface RMSD

The interface RMSD (iRMSD) is the backbone RMSD of the interface residues. Interface residues are defined as pairs of residues are defined as pairs of residues between protein and/or nucleic acid chains where any two heavy atoms are within 10Å of each other.

Text S6. Calculation of Z-TM-Comp

To re-rank prediction models for each oligomeric target in CASP/CAPRI, we introduce a cumulative Z-score metric termed Z-TM-Comp. This metric integrates three complementary similarity measures computed by TM-score-Comp: TM-score, rTM-score, and iTM-score, which respectively quantify global complex structural similarity, relative molecular orientation, and local interface similarity.

The Z-TM-Comp is computed following a protocol analogous to that used in CASP model assessment[5, 6]. Specifically, for each target:

1) Z-scores are calculated separately for TM-score, rTM-score, and iTM-score across the submitted models from all participating groups;
2) models with Z-scores below −2 for a given metric are excluded;
3) Z-scores are then recalculated for each metric using the reduced model set.

The final Z-TM-Comp for each model is defined as the average of the three recalculated Z-scores:

$$\text{Z-TM-Comp} = \frac{\text{Z-score(TM-score)} + \text{Z-score(rTM-score)} + \text{Z-score(iTM-score)}}{3} \tag{S3}$$

This cumulative score provides an integrated assessment of complex structure quality by jointly considering global topology, molecular orientation, and interface accuracy.

Supporting Figures

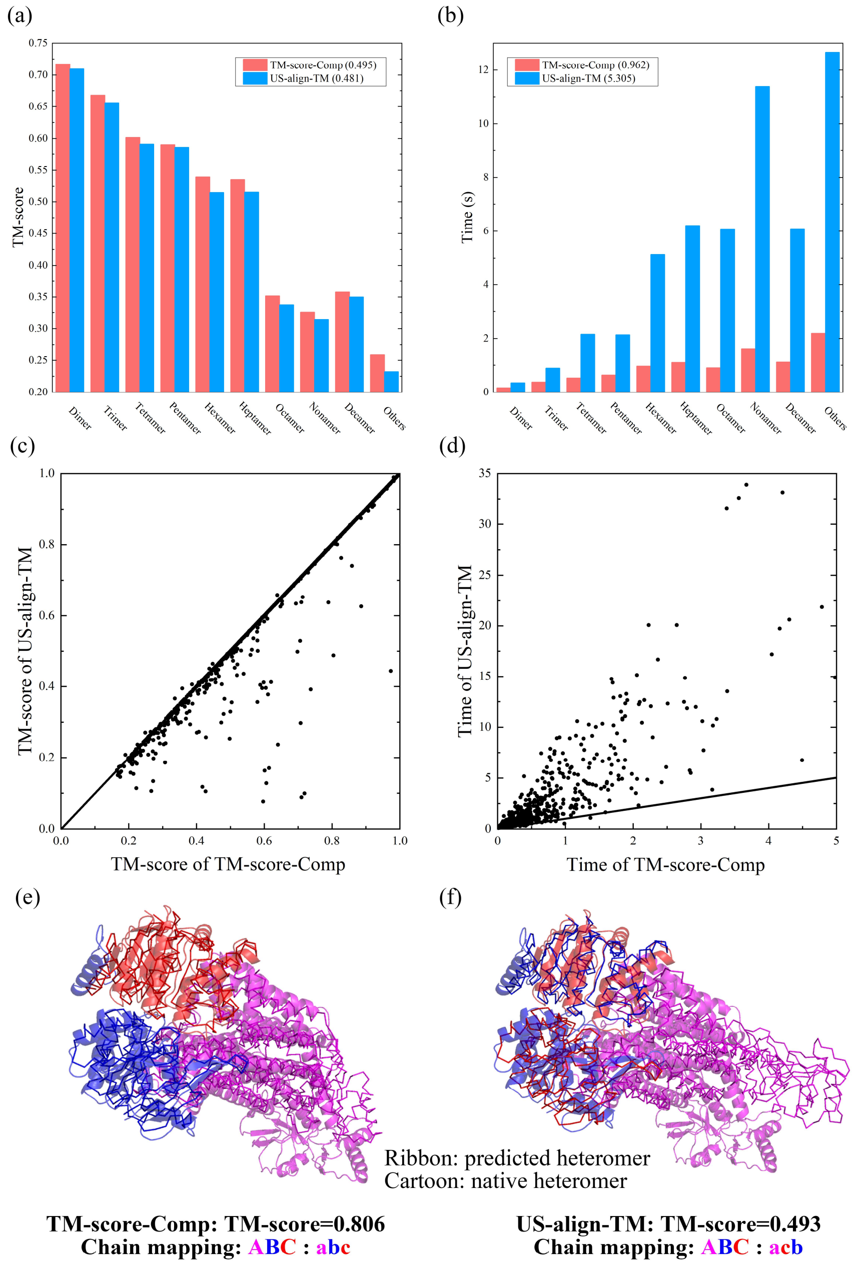

Fig. S1. Performance of two complex structure comparison methods on heteromeric protein complexes. a-b, Average TM-score (**a**) and running time (**b**) of TM-score-Comp and US-align-TM for heteromeric structure in different oligomer state (*x* axis) for n=811 dimers, 216 trimers, 223 tetramers, 20 pentamers, 68 hexamers, 16 heptamers, 20 octamers, 9 nonamers, 15 decamers and 22 others. The standard deviation is shown in Supplementary Table S2. **c-d,** Scatterplots for TM-score (**c**) and running time (**d**) of TM-score-Comp and US-align-TM. **e-f,** Structure comparison between the predicted (ribbon) and native (cartoon) complex structures of PDB 7TCH with stoichiometry of A2B1 by TM-score-Comp (**e**) and US-align-TM (**f**). Each chain of the heteromer is shown in different color.

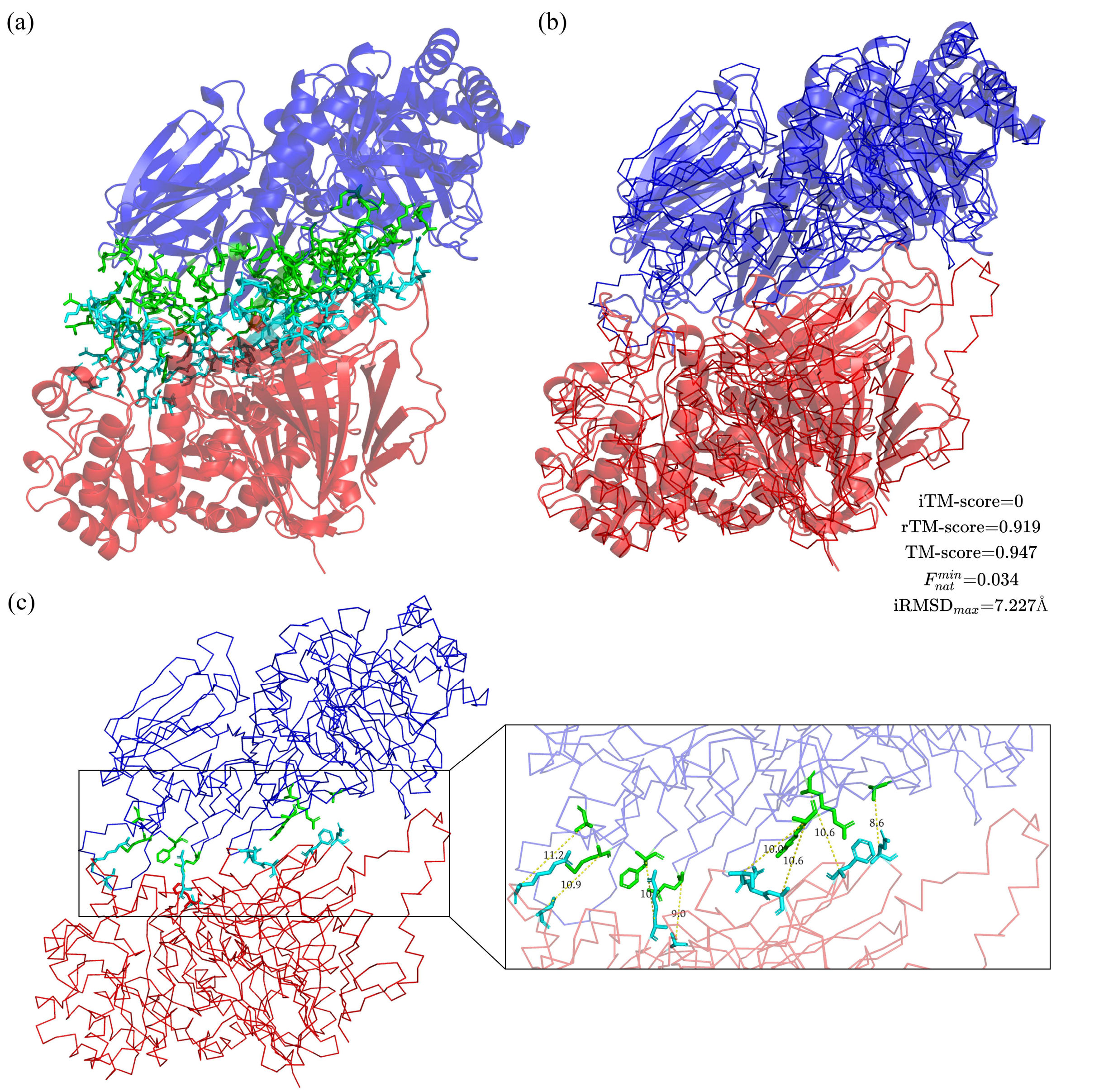

Fig. S2. Structure comparison between the predicted and native complex structures of PDB 2XN0. a. The native complex structure (cartoon) with interface residues (sticks). **b**. Superposition between the predicted (ribbon) and native (cartoon) complex structures. **c**. The predicted complex structure (ribbon) with the predicted interface, where the Ca distances of all predicted interface contacts are >8Å. Each chain of the complex and the interface residues in different chains are shown in different color.

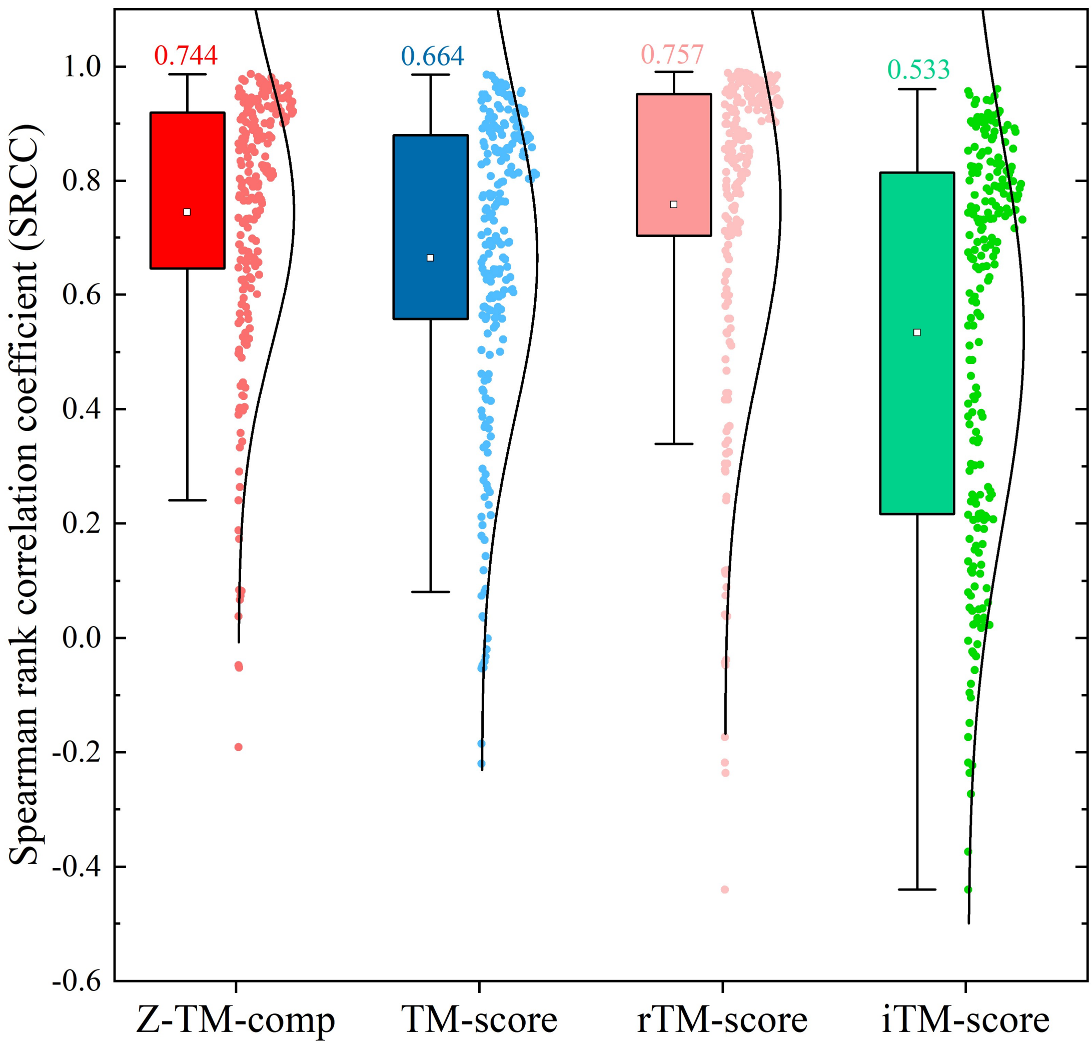

Fig. S3. Spearman rank correlation coefficient (SRCC) values of Z-TM-comp, TM-score, rTM-score, and iTM-score against DockQ.

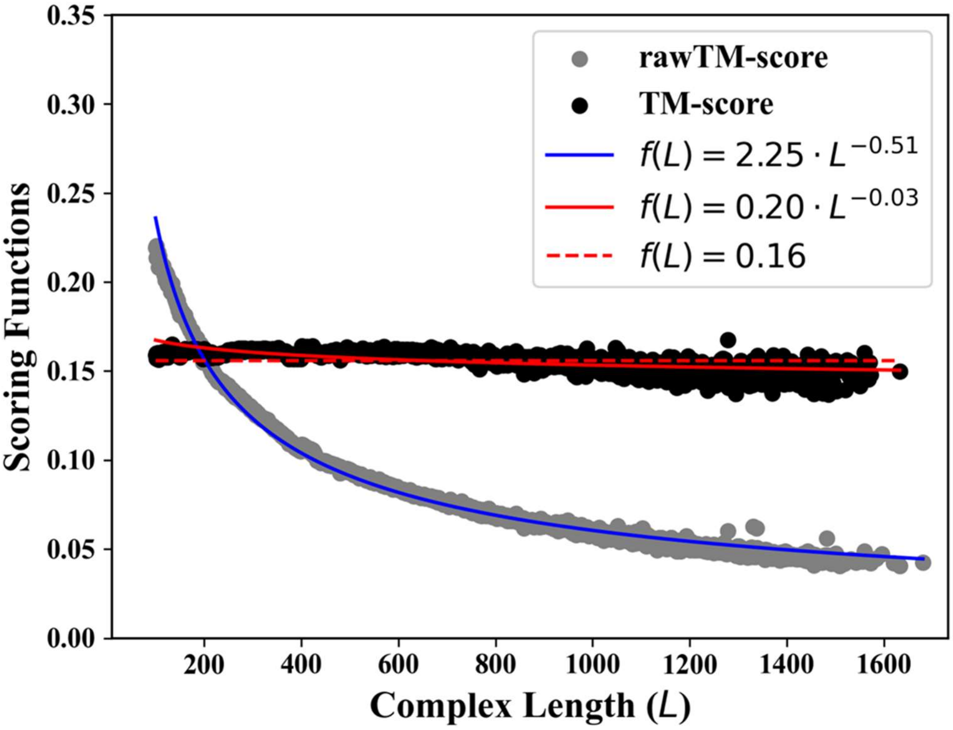

Fig. S4. The average TM-score and raw TM-score (rawTM-score) of random complex pairs as function of complex length, i.e., the sum of amino acid number in proteins, nucleotide number in DNA and RNA and heavy atom number in ligands. For the rawTM-score, d_0 for all molecule types are all equal to 5Å, while for the TM-score, the calculations of d_0 for different molecule types are defined as in **Eq. S1**. The data are calculated from all 3,000,000 non-redundant complex pairs which are randomly collected from PDB, consisting of 671,094 protein-protein, 1,875 protein-RNA, 4,716 protein-DNA, 2,314,355 protein-small molecule ligand, 2,009 protein-DNA-small molecule ligand, and 5,951 others complexes (such as protein-DNA-RNA and RNA-small molecule ligand complexes). For each complex pair, we only focus on their 3D coordinates and assume that the native indexes of residues for proteins, nucleotides for DNA/RNA and heavy atoms for ligands are matched between two complexes. The blue and red lines are the nonlinear least square Marquardt-Levenberg fits of the rawTM-score and TM-score data to a power-law equation $f(L)$, where L is the length of the smaller complex of the corresponding structure pairs. The red dashed line denotes the horizontal line of TM-score=0.16.

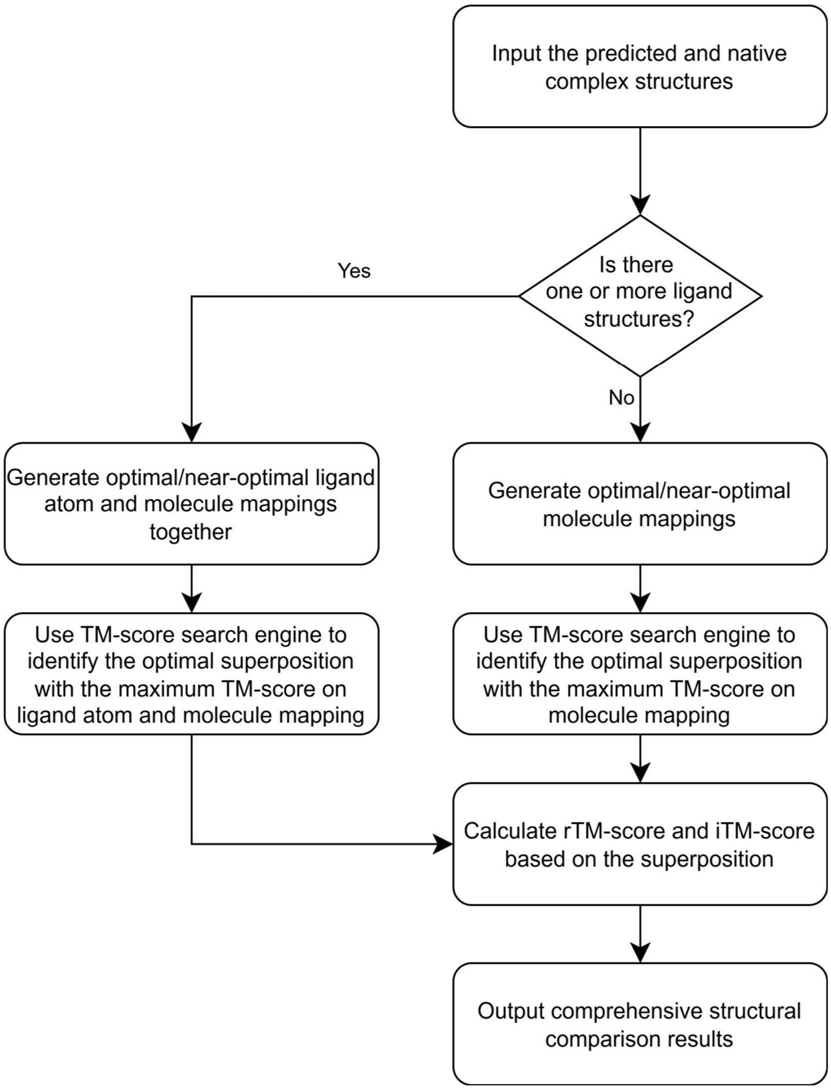

Fig. S5. **Pipeline of TM-score-Comp.**

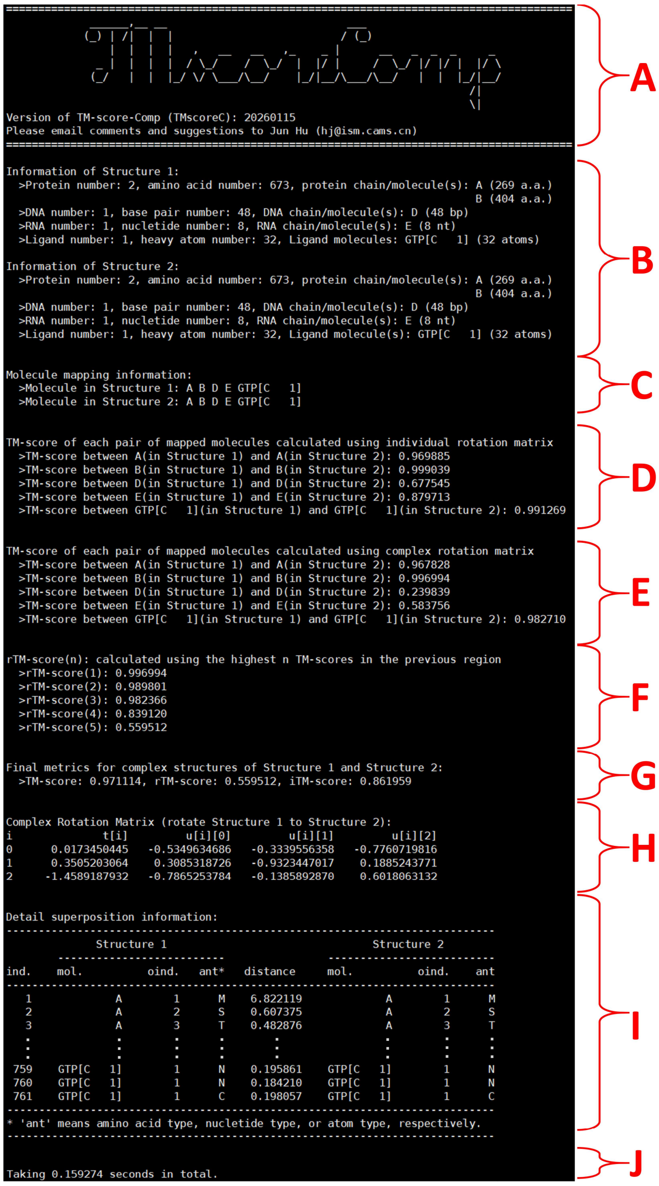

Fig. S6. Output example of the TM-score-Comp program. A. Program version and contact email. **B.** Inputted complex structure information. **C**. Molecule mapping result. **D-E**. single-molecule-level TM-scores calculated based on individual rotation matrix (**D**) and complex rotation matrix I. **F**. rTM-score(n) calculated using the highest n TM-scores calculated based on complex rotation matrix. In order to evaluate the orientation quality of prediction model which missed one or more molecules, we have also demonstrated the rTM-score with different levels. The first level rTM-score, i.e., rTM-score(1), is calculated based on the highest single-molecule-level TM-score, the second level rTM-score, rTM-score(2), is calculated based on the highest 2 single-molecule-level TM-scores, and so on. **G**. TM-score, rTM-score and iTM-score for superposition between two inputted complex structure. **H**. Complex rotation matrix. **I**. Distances of superimposed residues/nucleotides/ligand atoms. 'ind.' Is line indexes, 'mol.' Is molecule information, 'oind.' Is original index in inputted PDB file, 'ant' is amino acid type, nucleotide type or atom type, 'distance' is the distance between two superposed residues/nucleotides/atoms In proteins/DNAs (RNAs)/ligands.

Supporting Tables

Table S1. The average, standard deviation (SD) and *p*-values for TM-score and running time by TM-score-Comp and US-align-TM on HomoD.

Type	Methods	TM-score			Running time (s)		
		average	SD	*p*-value	average	SD	*p*-value
Dimer	TM-score-Comp	0.680	0.249	*	0.172	0.328	*
	US-align-TM	0.680	0.250	4.178e-09	0.622	1.233	1.945e-79
Trimer	TM-score-Comp	0.610	0.301	*	0.333	0.506	*
	US-align-TM	0.586	0.301	1.385e-05	1.566	2.564	6.272e-19
Tetramer	TM-score-Comp	0.538	0.258	*	0.410	0.602	*
	US-align-TM	0.526	0.264	1.110e-06	2.653	4.358	7.008e-15
Pentamer	TM-score-Comp	0.543	0.273	*	0.431	0.573	*
	US-align-TM	0.518	0.284	4.331e-04	2.955	4.212	3.194e-06
Hexamer	TM-score-Comp	0.498	0.282	*	0.703	0.938	*
	US-align-TM	0.489	0.288	5.492e-04	5.279	7.470	2.168e-06
Heptamer	TM-score-Comp	0.370	0.217	*	1.174	1.077	*
	US-align-TM	0.357	0.222	2.753e-03	8.746	8.146	5.956e-04
Octamer	TM-score-Comp	0.365	0.175	*	0.625	0.506	*
	US-align-TM	0.332	0.186	9.824e-05	4.629	4.847	1.075e-03
Nonamer	TM-score-Comp	0.294	0.274	*	1.007	0.890	*
	US-align-TM	0.282	0.280	1.409e-01	6.971	8.437	9.198e-02
Decamer	TM-score-Comp	0.363	0.204	*	1.031	1.131	*
	US-align-TM	0.356	0.208	5.292e-04	7.818	9.586	1.476e-04
Others	TM-score-Comp	0.274	0.179	*	1.111	1.032	*
	US-align-TM	0.258	0.183	4.178e-09	10.106	10.807	1.945e-79

* The *p*-values are from two-tailed paired Student’s t-tests against TM-score-Comp.

Table S2. The average, standard deviation (SD) and *p*-values for TM-score and running time by TM-score-Comp and US-align-TM on HeteroD.

Types	Methods	TM-score			Running time (s)		
		average	SD	*p*-value	average	SD	*p*-value
Dimer	TM-score-Comp	0.717	0.198	*	0.161	0.416	*
	US-align-TM	0.709	0.210	6.656e-05	0.351	0.894	2.571e-24
Trimer	TM-score-Comp	0.669	0.199	*	0.376	0.532	*
	US-align-TM	0.656	0.202	1.623e-03	0.890	1.491	3.747e-11
Tetramer	TM-score-Comp	0.604	0.234	*	0.527	0.677	*
	US-align-TM	0.593	0.240	1.235e-04	2.156	2.898	1.705e-21
Pentamer	TM-score-Comp	0.593	0.233	*	0.638	0.513	*
	US-align-TM	0.587	0.237	1.710e-02	2.140	1.878	2.018e-04
Hexamer	TM-score-Comp	0.543	0.234	*	0.966	1.275	*
	US-align-TM	0.518	0.244	1.498e-04	5.132	7.117	3.410e-07
Heptamer	TM-score-Comp	0.514	0.267	*	1.104	0.941	*
	US-align-TM	0.492	0.279	2.349e-02	6.204	4.225	3.124e-05
Octamer	TM-score-Comp	0.348	0.089	*	0.909	0.728	*
	US-align-TM	0.333	0.092	1.649e-01	6.060	5.196	8.690e-05
Nonamer	TM-score-Comp	0.326	0.120	*	1.621	1.324	*
	US-align-TM	0.314	0.110	5.652e-02	11.389	11.552	2.750e-02
Decamer	TM-score-Comp	0.358	0.195	*	1.121	0.855	*
	US-align-TM	0.350	0.197	2.839e-05	6.068	5.676	3.482e-04
Others	TM-score-Comp	0.259	0.071	*	2.195	2.160	*
	US-align-TM	0.232	0.069	6.656e-05	12.664	12.956	2.571e-24

* The *p*-values are from two-tailed paired Student's t-tests against TM-score-Comp.

Table S3. TM-score and running time (RT) values by TM-score-Comp and US-align-TM (option '-TM-score' of US-align) on 114 non-redundant complex targets where each target composed of proteins and DNAs/RNAs.

Targets	TM-score-Comp		US-align-TM		Targets	TM-score-Comp		US-align-TM	
	TM-score	RT (s)	TM-score	RT (s)		TM-score	RT (s)	TM-score	RT (s)
1hys	0.696	0.003	0.696	3.820	7v9x	0.751	0.000	0.637	1.950
3e2e	0.985	0.341	0.982	1.560	7wb0	0.903	0.041	0.903	2.440
4gg4	0.986	0.176	0.987	0.440	7wb1	0.933	0.054	0.894	3.120
4h8k	0.987	0.286	0.987	0.170	7wju	0.852	0.058	0.822	1.380
4hkq	0.987	0.309	0.987	0.330	7wwv	0.682	0.002	0.682	36.990
5c44	0.992	0.043	0.992	26.000	7yhs	0.880	0.004	0.880	46.130
5kk5	0.988	0.122	0.988	2.170	7yoj	0.769	0.052	0.769	4.490
5o6u	0.988	0.224	0.988	3.130	7z4c	0.824	0.059	0.824	11.320
5u0a	0.968	0.215	0.968	29.270	7z4e	0.860	0.139	0.860	7.340
6aeg	0.950	0.074	0.950	5.480	7z4g	0.930	0.142	0.929	5.930
6b44	0.865	0.006	0.865	42.840	7z4h	0.878	0.119	0.878	7.900
6gte	0.851	0.052	0.851	9.790	7z4k	0.878	0.084	0.876	8.640
6k3z	0.998	0.150	0.997	2.610	8axa	0.782	0.069	0.759	1.380
6k4s	0.968	0.157	0.968	4.660	8ctl	0.784	0.123	0.784	0.980
6k4u	0.973	0.061	0.973	4.030	8dc2	0.948	0.029	0.903	1.280
6k57	0.998	0.257	0.998	2.590	8dej	0.823	0.006	0.823	36.040
6kc7	0.795	0.203	0.794	3.980	8ex9	0.754	0.107	0.769	0.270
6kc8	0.962	0.175	0.962	3.810	8fcj	0.909	0.001	0.899	25.570
6lnb	0.759	0.006	0.760	35.370	8fd3	0.912	0.000	0.910	56.180
6lu0	0.984	0.104	0.987	1.830	8fzt	0.916	0.080	0.916	6.160
6ne0	0.901	0.019	0.902	43.010	8g1i	0.919	0.051	0.837	6.030
6o0y	0.883	0.162	0.881	3.440	8g5p	0.860	0.001	0.859	11.160
6o0z	0.873	0.081	0.791	6.990	8gam	0.941	0.003	0.940	27.130
6pij	0.768	0.002	0.768	55.510	8gan	0.804	0.000	0.804	57.760
6rj9	0.682	0.003	0.678	3.030	8h67	0.849	0.000	0.846	38.920
6rja	0.409	0.000	0.406	10.640	8h7q	0.884	0.000	0.884	54.050
6rjd	0.925	0.053	0.920	1.870	8hnv	0.820	0.002	0.819	3.460
6rjg	0.952	0.098	0.949	3.080	8hnw	0.977	0.170	0.976	1.700
6tye	0.993	0.365	0.993	15.060	8hr5	0.802	0.013	0.784	2.550
6tyf	0.993	0.225	0.993	15.340	8hud	0.950	0.030	0.950	1.520
6tyg	0.994	0.217	0.994	15.240	8i54	0.782	0.012	0.782	4.920
6w5c	0.983	0.110	0.983	2.220	8iew	0.866	0.021	0.875	0.840
6w64	0.969	0.120	0.969	1.660	8inb	0.868	0.043	0.877	0.770
6whi	0.870	0.000	0.853	52.640	8isz	0.924	0.001	0.924	1.700
7bg9	0.752	0.002	0.751	5.870	8it0	0.875	0.001	0.875	9.300
7ecw	0.861	0.002	0.861	31.640	8it1	0.593	0.001	0.593	71.830
7el1	0.830	0.002	0.767	3.780	8iyq	0.893	0.010	0.893	5.440
7eqg	0.905	0.034	0.905	36.550	8jg9	0.402	0.000	0.371	24.580
7lys	0.983	0.161	0.907	1.010	8jtj	0.950	0.054	0.950	2.780
7n3p	0.739	0.055	0.739	1.360	8kam	0.946	0.056	0.946	6.270
7o0g	0.782	0.079	0.783	0.850	8pm4	0.933	0.001	0.932	1.090
7pla	0.778	0.068	0.738	1.240	8qg0	0.936	0.042	0.936	1.120
7qxa	0.790	0.003	0.783	7.540	8sfh	0.758	0.145	0.758	7.960
7qxs	0.689	0.002	0.683	11.320	8sfi	0.817	0.182	0.817	9.970
7s4u	0.911	0.020	0.911	5.070	8sfj	0.904	0.202	0.904	4.410
7s4v	0.924	0.151	0.924	3.930	8sfn	0.972	0.243	0.972	3.680
7sba	0.982	0.001	0.982	27.100	8sfr	0.923	0.159	0.923	8.990
7tax	0.977	0.005	0.977	30.220	8t6o	0.825	0.000	0.822	6.610
7trd	0.931	0.005	0.881	2.450	8t6s	0.934	0.005	0.934	2.510
7tre	0.931	0.003	0.887	3.790	8t6t	0.942	0.025	0.942	4.730
7trf	0.803	0.001	0.760	4.030	8t6y	0.990	0.019	0.990	2.500
7u5e	0.917	0.003	0.917	34.370	8t76	0.950	0.046	0.950	3.030
7utn	0.785	0.098	0.785	0.930	8t77	0.897	0.008	0.897	7.480
7v59	0.724	0.121	0.723	39.910	8t78	0.967	0.005	0.966	1.810
7v94	0.928	0.001	0.900	3.220	8uza	0.919	0.115	0.914	2.990
7v99	0.795	0.013	0.792	4.480	8uzb	0.945	0.103	0.940	2.150
7v9u	0.764	0.000	0.749	1.160	8wmh	0.955	0.022	0.939	3.920

Table S4. Classification performance comparison of rTM-score, TM-score and rLRMSD on protein-ligand docking pose dataset.

Metrics	Sen	Spe	Acc	Pre	F1	MCC	AUC	AUCPR
rTM-score	0.997	1.000	0.999	1.000	0.999	0.998	1.000	0.997
TM-score	0.728	0.827	0.807	0.526	0.611	0.498	0.860	0.670
rLRMSD	0.817	0.924	0.902	0.741	0.777	0.716	0.945	0.780

Table S5. The predicted structure numbers (PSN) of all 224 CASP/CAPRI targets

Target	PSN	Target	PSN	Target	PSN	Target	PSN	Target	PSN
H0953	62	T037.1	390	T073.1	221	T1123o	242	T136.3	288
H0957	61	T037.2	390	T073.2	221	T1124o	270	T138.1	218
H0968	63	T038.1	399	T075.1	253	T1127o	279	T138.2	265
H1036	135	T039.1	366	T077.1	229	T113.1	254	T140.1	279
H1072	119	T040.1	368	T078.1	229	T1132o	224	T141.1	163
H1106	308	T040.2	368	T079.1	242	T1153o	221	T142.1	270
H1111	225	T041.1	327	T080.1	264	T1160o	261	T143.1	194
H1114	275	T042.1	278	T081.1	206	T1161o	266	T144.1	240
H1114v2	275	T046.1	387	T085.1	241	T116.1	295	T146.1	267
H1129	299	T047.1	277	T086.1	251	T1170o	257	T146.2	281
H1134	319	T050.1	354	T086.2	251	T117.1	251	T146.3	281
H1135	307	T053.1	384	T087.1	231	T117.2	231	T149.1	222
H1137	203	T054.1	377	T088.1	261	T1173o	286	T149.2	222
H1140	316	T058.1	230	T089.1	211	T117.3	251	T149.3	217
H1141	312	T060.1	320	T090.1	221	T1174o	278	T149.4	222
H1142	288	T060.2	320	T091.1	242	T117.4	236	T149.5	202
H1143	305	T060.3	320	T092.1	225	T1176o	254	T150.1	210
H1144	313	T061.1	320	T093.1	213	T1176v1o	223	T150.2	210
H1151	316	T061.2	320	T094.1	215	T1178o	265	T150.3	210
H1157	304	T061.3	320	T095.1	250	T1179o	265	T150.4	210
H1166	291	T062.1	310	T0960o	35	T1181o	276	T150.5	190
H1166v1	291	T062.2	320	T096.1	270	T1187o	292	T151.1	183
H1167	306	T062.3	310	T0966o	59	T119.1	270	T151.2	183
H1167v1	306	T063.1	314	T097.1	270	T1192o	211	T151.3	193
H1168	303	T063.2	314	T098.1	268	T1201o	361	T151.4	193
H1168v1	303	T063.3	314	T099.1	250	T120.1	276	T151.5	173
H1185	254	T064.1	318	T100.1	240	T1206o	357	T152.1	292
H1202	375	T064.2	318	T1003o	77	T121.1	327	T153.1	305
H1204	367	T064.3	318	T1009o	76	T122.1	339	T154.1	245
H1213	352	T065.1	356	T101.1	248	T123.1	320	T158.1	238
H1222	376	T065.2	356	T1016o	76	T1234o	357	T159.1	203
H1223	373	T066.1	352	T1032o	128	T1235o	360	T159.2	109
H1225	377	T066.2	352	T1038o	74	T1237o	344	T159.3	81
H1232	363	T066.3	352	T104.1	326	T1240o	348	T159.4	40
H1233	342	T067.1	351	T105.1	337	T124.1	290	T159.5	40
H1236	347	T067.2	351	T1054o	151	T125.1	282	T159.6	111
H1272	253	T068.1	221	T107.1	250	T125.2	291	T159.7	133
T024.1	335	T068.2	221	T1073o	148	T125.3	291	T164.1	243
T025.1	336	T069.1	256	T1080o	80	T125.4	291	T165.1	212
T026.1	351	T070.1	221	T110.1	235	T1298o	331	T166.1	223
T029.1	381	T070.2	221	T1109o	243	T131.1	289	T169.1	260
T030.1	350	T071.1	212	T1110o	241	T132.1	299	T176.1	248
T032.1	354	T071.2	212	T1113o	233	T133.1	343	T180.1	201
T035.1	339	T071.3	212	T1115o	187	T136.1	288	T180.2	201
T036.1	319	T072.1	234	T1121o	231	T136.2	288		

Table S6. Spearman rank correlation coefficient (SRCC) values between the orders of predicted models ranked by CASP-score with those ranked by Z-TM-comp, TM-score, rTM-score, and iTM-score on each CASP/CAPRI target

Target	Z-TM-comp	TM-score	rTM-score	iTM-score
H0953	0.778	0.607	0.295	0.574
H0957	0.792	0.639	0.326	0.208
H0968	0.775	0.863	0.568	0.265
H1036	0.789	0.751	0.391	0.848
H1072	0.782	0.812	0.453	0.503
H1106	0.899	0.896	0.900	0.844
H1111	0.805	0.805	0.758	0.793
H1114	0.869	0.833	0.706	0.765
H1114v2	0.650	0.679	0.206	0.318
H1129	0.900	0.897	0.885	0.687
H1134	0.962	0.947	0.936	0.941
H1135	0.822	0.752	0.811	0.633
H1137	0.916	0.961	0.722	0.764
H1140	0.801	0.812	0.724	0.392
H1141	0.877	0.883	0.839	0.702
H1142	0.679	0.710	0.397	0.138
H1143	0.965	0.979	0.976	0.939
H1144	0.926	0.951	0.871	0.772
H1151	0.809	0.815	0.798	0.767
H1157	0.723	0.652	0.635	0.588
H1166	0.752	0.616	0.395	0.736
H1166v1	0.559	0.167	0.545	0.375
H1167	0.673	0.546	0.620	0.724
H1167v1	0.561	0.523	0.339	-0.067
H1168	0.939	0.901	0.910	0.895
H1168v1	0.778	0.744	0.782	0.737
H1185	0.616	0.714	0.630	0.560
H1202	0.817	0.858	0.800	0.646
H1204	0.654	0.685	0.525	0.533
H1213	0.879	0.852	0.868	0.855
H1222	0.901	0.626	0.856	0.902
H1223	0.667	0.612	0.130	0.594
H1225	0.767	0.776	-0.048	0.736
H1232	0.796	0.705	0.643	0.764
H1233	0.888	0.874	0.809	0.894
H1236	0.628	0.599	0.456	0.042
H1272	0.798	0.849	0.434	0.661
T024.1	0.842	0.863	0.748	0.311
T025.1	0.875	0.905	0.813	0.509
T026.1	0.924	0.897	0.797	0.558
T029.1	0.843	0.851	0.683	0.255
T030.1	0.821	0.834	0.749	0.224
T032.1	0.838	0.821	0.769	0.745
T035.1	0.874	0.829	0.641	0.180
T036.1	0.848	0.779	0.700	0.011

T037.1	0.673	0.681	0.553	0.708
T037.2	0.674	0.681	0.553	0.708
T038.1	0.668	0.655	0.497	0.038
T039.1	0.741	0.741	0.428	0.197
T040.1	0.834	0.857	0.799	0.640
T040.2	0.860	0.888	0.804	0.718
T041.1	0.945	0.943	0.941	0.898
T042.1	0.850	0.887	0.603	0.254
T046.1	0.878	0.843	0.351	0.478
T047.1	0.954	0.916	0.939	0.894
T050.1	0.797	0.822	0.604	0.620
T053.1	0.938	0.930	0.893	0.374
T054.1	0.843	0.805	0.745	0.508
T058.1	0.890	0.901	0.842	0.710
T060.1	0.958	0.919	0.661	0.719
T060.2	0.951	0.929	0.745	0.754
T060.3	0.933	0.911	0.732	0.745
T061.1	0.925	0.867	0.710	0.549
T061.2	0.915	0.899	0.833	0.798
T061.3	0.914	0.895	0.823	0.730
T062.1	0.948	0.894	0.686	0.570
T062.2	0.911	0.907	0.721	0.642
T062.3	0.924	0.919	0.740	0.742
T063.1	0.873	0.822	0.697	0.589
T063.2	0.920	0.901	0.863	0.793
T063.3	0.912	0.891	0.860	0.763
T064.1	0.961	0.906	0.685	0.634
T064.2	0.884	0.868	0.800	0.792
T064.3	0.882	0.846	0.798	0.746
T065.1	0.884	0.866	0.211	0.080
T065.2	0.833	0.547	0.454	0.262
T066.1	0.680	0.680	0.680	0.015
T066.2	0.477	0.686	-0.398	-0.181
T066.3	0.657	0.657	0.657	0.017
T067.1	0.947	0.904	0.950	0.912
T067.2	0.784	0.784	-0.313	-0.313
T068.1	0.637	0.695	0.492	0.101
T068.2	0.634	0.640	0.489	-0.060
T069.1	0.963	0.971	0.901	0.803
T070.1	0.891	0.769	0.550	0.320
T070.2	0.940	0.933	0.653	0.492
T071.1	0.920	0.862	0.747	0.078
T071.2	0.737	0.644	0.549	0.056
T071.3	0.899	0.841	0.788	-0.071
T072.1	0.918	0.913	0.846	0.621
T073.1	0.851	0.841	0.729	0.179
T073.2	0.690	0.699	0.526	0.156
T075.1	0.881	0.931	0.801	0.653
T077.1	0.746	0.541	0.516	0.310

T078.1	0.925	0.885	0.735	0.775
T079.1	0.929	0.905	0.785	0.171
T080.1	0.971	0.967	0.945	0.881
T081.1	0.469	0.383	0.628	0.626
T085.1	0.948	0.939	0.881	0.770
T086.1	0.923	0.913	0.730	0.688
T086.2	0.922	0.903	0.535	0.187
T087.1	0.945	0.953	0.914	0.885
T088.1	0.838	0.817	0.695	0.477
T089.1	0.957	0.948	0.908	0.851
T090.1	0.953	0.941	0.930	0.863
T091.1	0.947	0.934	0.918	0.911
T092.1	0.972	0.967	0.952	0.919
T093.1	0.922	0.926	0.920	0.849
T094.1	0.885	0.891	0.842	0.750
T095.1	0.543	0.312	0.173	0.410
T0960o	0.915	0.857	0.938	0.862
T096.1	0.864	0.847	0.823	-0.151
T0966o	0.829	0.554	0.577	-0.262
T097.1	0.844	0.830	0.842	0.274
T098.1	0.873	0.810	0.794	0.368
T099.1	0.849	0.849	0.742	0.535
T100.1	0.751	0.756	0.555	0.527
T1003o	0.903	0.819	0.846	0.937
T1009o	0.940	0.951	0.780	0.568
T101.1	0.612	0.554	0.404	0.224
T1016o	0.922	0.818	0.827	0.836
T1032o	0.922	0.907	0.891	0.875
T1038o	0.570	0.632	0.389	0.010
T104.1	0.966	0.970	0.965	0.906
T105.1	0.938	0.948	0.946	0.907
T1054o	0.838	0.765	0.396	0.136
T107.1	0.738	0.665	0.816	0.187
T1073o	0.735	0.638	0.565	0.266
T1080o	0.933	0.891	0.898	0.672
T110.1	0.974	0.978	0.960	0.911
T1109o	0.833	0.853	0.850	0.660
T1110o	0.793	0.861	0.872	0.729
T1113o	0.874	0.771	0.758	0.840
T1115o	0.716	0.700	0.533	0.788
T1121o	0.396	0.431	0.309	0.195
T1123o	0.968	0.978	0.940	0.927
T1124o	0.904	0.848	0.831	0.823
T1127o	0.910	0.876	0.885	0.858
T113.1	0.930	0.793	0.702	0.431
T1132o	0.887	0.912	0.908	0.815
T1153o	0.905	0.866	0.860	0.769
T1160o	0.183	0.048	0.172	0.433
T1161o	0.834	0.845	0.850	0.660

T116.1	0.867	0.720	0.731	0.849
T1170o	0.891	0.761	0.788	0.741
T117.1	0.931	0.769	0.824	0.729
T117.2	0.794	0.892	0.491	0.248
T1173o	0.730	0.866	0.856	0.536
T117.3	0.930	0.782	0.820	0.759
T1174o	0.923	0.934	0.913	0.826
T117.4	0.814	0.825	0.442	0.167
T1176o	0.770	0.742	0.575	0.343
T1176v1o	0.370	0.268	0.374	0.285
T1178o	0.961	0.974	0.960	0.939
T1179o	0.970	0.944	0.923	0.891
T1181o	0.958	0.906	0.940	0.894
T1187o	0.863	0.859	0.834	0.611
T119.1	0.958	0.947	0.921	0.904
T1192o	0.773	0.698	0.436	0.838
T1201o	0.684	0.655	0.669	0.659
T120.1	0.943	0.934	0.937	0.770
T1206o	0.868	0.896	0.897	0.618
T121.1	0.837	0.499	0.725	0.693
T122.1	0.905	0.886	0.528	0.717
T123.1	0.798	0.807	0.546	0.190
T1234o	0.912	0.936	0.895	0.868
T1235o	0.867	0.928	0.796	0.855
T1237o	0.431	0.420	0.437	0.340
T1240o	0.592	0.506	0.503	0.552
T124.1	0.825	0.851	0.326	0.337
T125.1	0.935	0.933	0.928	0.836
T125.2	0.712	0.701	0.708	0.664
T125.3	0.826	0.826	0.735	0.025
T125.4	0.717	0.703	0.475	0.101
T1298o	0.774	0.738	0.726	0.702
T131.1	0.677	0.638	0.613	0.311
T132.1	0.673	0.539	0.603	0.190
T133.1	0.958	0.959	0.927	0.783
T136.1	0.859	0.802	0.840	0.792
T136.2	0.964	0.854	0.868	0.898
T136.3	0.825	0.875	0.894	0.111
T138.1	0.933	0.874	0.675	0.335
T138.2	0.920	0.938	0.495	0.301
T140.1	0.988	0.984	0.964	0.951
T141.1	0.914	0.927	0.626	0.648
T142.1	0.963	0.949	0.796	0.639
T143.1	0.848	0.869	0.860	0.732
T144.1	0.950	0.884	0.889	0.841
T146.1	0.884	0.885	0.778	0.800
T146.2	0.842	0.729	0.715	0.306
T146.3	0.816	0.777	0.284	0.157
T149.1	0.952	0.940	0.860	0.839

T149.2	0.861	0.881	0.531	0.107
T149.3	0.923	0.865	0.839	0.844
T149.4	0.830	0.906	0.372	0.031
T149.5	0.873	0.819	0.684	0.080
T150.1	0.873	0.881	0.773	0.687
T150.2	0.910	0.948	0.575	0.466
T150.3	0.864	0.785	0.773	0.721
T150.4	0.853	0.788	0.719	0.204
T150.5	0.856	0.597	0.816	0.078
T151.1	0.940	0.906	0.835	0.870
T151.2	0.805	0.876	0.452	-0.037
T151.3	0.912	0.790	0.767	0.709
T151.4	0.918	0.884	0.696	0.176
T151.5	0.595	0.590	0.331	0.281
T152.1	0.962	0.928	0.932	0.957
T153.1	0.959	0.964	0.940	0.822
T154.1	0.955	0.959	0.827	0.493
T158.1	0.943	0.942	0.924	0.880
T159.1	0.978	0.963	0.965	0.966
T159.2	0.965	0.871	0.890	0.793
T159.3	0.847	0.696	0.820	0.410
T159.4	0.741	0.845	0.635	0.018
T159.5	0.657	0.595	0.702	0.069
T159.6	0.853	0.719	0.690	0.872
T159.7	0.908	0.860	0.836	0.546
T164.1	0.923	0.879	0.879	0.820
T165.1	0.484	0.484	0.116	0.659
T166.1	0.962	0.967	0.949	0.839
T169.1	0.778	0.670	0.325	-0.056
T176.1	0.864	0.810	0.488	0.234
T180.1	0.943	0.945	0.911	0.617
T180.2	0.946	0.865	0.878	0.903
Average	0.836	0.805	0.695	0.550

Table S7. Spearman rank correlation coefficient (SRCC) values between the orders of predicted models ranked by DockQ with those ranked by Z-TM-comp, TM-score, rTM-score, and iTM-score on each CASP/CAPRI target

Target	Z-TM-comp	Z-score(TM-score) +Z-score(rTM-score)	Z-score(TM-score) +Z-score(iTM-score)	Z-score(rTM-score) +Z-score(iTM-score)	TM-score	rTM-score	iTM-score
H0953	0.497	0.475	0.356	0.318	0.211	0.305	0.743
H0957	0.550	0.396	0.356	0.605	0.178	0.416	0.080
H0968	0.390	0.408	0.418	0.109	0.462	0.117	0.134
H1036	0.734	0.658	0.741	0.721	0.704	0.487	0.704
H1072	0.499	0.484	0.449	0.510	0.434	0.339	0.409
H1106	0.947	0.935	0.909	0.931	0.940	0.933	0.852
H1111	0.770	0.770	0.770	-0.218	0.770	-0.218	-0.218
H1114	0.774	0.774	0.774	-0.174	0.774	-0.174	-0.174
H1114v2	0.504	0.580	0.494	0.073	0.657	0.040	0.052
H1129	0.962	0.955	0.965	0.966	0.951	0.954	0.754
H1134	0.955	0.919	0.967	0.963	0.926	0.913	0.957
H1135	0.648	0.675	0.616	0.116	0.646	0.112	0.073
H1137	0.907	0.904	0.925	0.749	0.951	0.711	0.745
H1140	0.865	0.861	0.836	0.914	0.828	0.913	0.387
H1141	0.954	0.950	0.948	0.960	0.947	0.960	0.740
H1142	0.567	0.567	0.504	0.737	0.503	0.737	-0.005
H1143	0.978	0.987	0.974	0.976	0.986	0.989	0.950
H1144	0.913	0.910	0.891	0.945	0.891	0.954	0.774
H1151	0.853	0.858	0.839	0.845	0.867	0.855	0.800
H1157	0.704	0.651	0.652	0.605	0.733	0.599	0.486
H1166	0.936	0.851	0.885	0.792	0.860	0.322	0.825
H1166v1	0.688	0.496	0.337	0.842	0.073	0.791	0.458
H1167	0.803	0.704	0.940	0.634	0.899	0.428	0.904
H1167v1	0.687	0.671	0.683	0.407	0.664	0.361	-0.096
H1168	0.975	0.952	0.973	0.968	0.944	0.955	0.901
H1168v1	0.946	0.929	0.937	0.925	0.909	0.949	0.813
H1185	0.834	0.803	0.795	0.820	0.833	0.785	0.751
H1202	0.827	0.810	0.758	0.794	0.847	0.775	0.674
H1204	0.902	0.847	0.918	0.873	0.893	0.467	0.824
H1213	0.919	0.898	0.915	0.917	0.885	0.899	0.903
H1222	0.874	0.847	0.805	0.926	0.564	0.927	0.859
H1223	0.789	0.794	0.764	0.732	0.748	-0.043	0.732
H1225	0.797	0.747	0.705	0.631	0.808	0.118	0.603
H1232	0.725	0.623	0.781	0.764	0.638	0.580	0.726
H1233	0.888	0.875	0.890	0.891	0.881	0.815	0.894
H1236	0.398	0.317	0.485	0.302	0.296	0.294	0.173
H1272	0.862	0.835	0.747	0.717	0.876	0.599	0.546
T024.1	0.764	0.733	0.712	0.818	0.687	0.800	0.373
T025.1	0.927	0.928	0.900	0.955	0.897	0.959	0.511
T026.1	0.871	0.861	0.861	0.973	0.751	0.978	0.665
T029.1	0.876	0.875	0.859	0.968	0.857	0.966	0.304
T030.1	0.882	0.878	0.859	0.921	0.851	0.920	0.394
T032.1	0.936	0.898	0.941	0.941	0.892	0.899	0.731
T035.1	0.555	0.553	0.435	0.736	0.397	0.732	0.118
T036.1	0.661	0.659	0.450	0.705	0.431	0.702	0.114
T037.1	0.583	0.573	0.588	0.577	0.578	0.534	0.663
T037.2	0.583	0.574	0.588	0.575	0.579	0.532	0.662
T038.1	0.626	0.623	0.571	0.839	0.557	0.839	0.125
T039.1	0.677	0.652	0.628	0.955	0.627	0.955	0.154
T040.1	0.936	0.925	0.934	0.934	0.920	0.923	0.713
T040.2	0.933	0.928	0.920	0.927	0.908	0.925	0.717
T041.1	0.987	0.985	0.985	0.984	0.984	0.983	0.934
T042.1	0.782	0.782	0.637	0.964	0.637	0.964	0.215
T046.1	0.609	0.599	0.406	0.717	0.386	0.709	0.292
T047.1	0.945	0.878	0.947	0.961	0.849	0.919	0.941
T050.1	0.919	0.881	0.903	0.966	0.852	0.907	0.559
T053.1	0.911	0.917	0.887	0.925	0.889	0.937	0.345
T054.1	0.778	0.757	0.724	0.862	0.672	0.851	0.486
T058.1	0.940	0.933	0.922	0.947	0.911	0.945	0.678
T060.1	0.627	0.627	0.621	0.991	0.561	0.991	0.731
T060.2	0.730	0.715	0.700	0.971	0.694	0.971	0.763
T060.3	0.677	0.670	0.655	0.987	0.623	0.987	0.786
T061.1	0.618	0.614	0.607	0.984	0.532	0.985	0.743
T061.2	0.732	0.730	0.725	0.986	0.710	0.986	0.781
T061.3	0.717	0.710	0.704	0.990	0.685	0.990	0.813
T062.1	0.656	0.655	0.649	0.978	0.574	0.979	0.649

T062.2	0.644	0.639	0.634	0.935	0.635	0.936	0.732
T062.3	0.665	0.662	0.656	0.988	0.645	0.988	0.763
T063.1	0.526	0.524	0.521	0.985	0.450	0.987	0.733
T063.2	0.764	0.749	0.750	0.983	0.736	0.985	0.824
T063.3	0.733	0.723	0.717	0.987	0.697	0.988	0.844
T064.1	0.675	0.674	0.669	0.988	0.593	0.988	0.589
T064.2	0.669	0.658	0.652	0.973	0.644	0.973	0.846
T064.3	0.631	0.628	0.619	0.989	0.572	0.989	0.850
T065.1	-0.191	-0.191	-0.194	0.074	-0.220	0.074	0.047
T065.2	0.240	0.184	0.057	0.587	-0.053	0.587	0.238
T066.1	-0.048	-0.048	-0.048	-0.048	-0.048	-0.048	0.023
T066.2	0.402	0.392	0.227	0.770	0.197	0.770	0.090
T066.3	0.037	0.037	0.037	0.037	0.037	0.037	-0.149
T067.1	0.905	0.889	0.871	0.954	0.834	0.960	0.911
T067.2	0.333	0.333	0.333	-0.236	0.333	-0.236	-0.236
T068.1	0.612	0.605	0.607	0.568	0.601	0.558	0.161
T068.2	0.358	0.339	0.307	0.315	0.276	0.291	0.026
T069.1	0.941	0.941	0.913	0.986	0.911	0.988	0.791
T070.1	0.594	0.585	0.460	0.870	0.368	0.864	0.192
T070.2	0.851	0.847	0.705	0.887	0.694	0.883	0.546
T071.1	0.841	0.841	0.615	0.946	0.599	0.950	0.149
T071.2	0.628	0.607	0.396	0.787	0.373	0.769	0.034
T071.3	0.815	0.808	0.645	0.927	0.637	0.924	-0.080
T072.1	0.938	0.935	0.882	0.939	0.880	0.947	0.597
T073.1	0.800	0.848	0.716	0.794	0.777	0.857	0.112
T073.2	0.698	0.715	0.577	0.703	0.594	0.720	0.049
T075.1	0.943	0.940	0.857	0.988	0.851	0.986	0.692
T077.1	0.398	0.330	0.259	0.658	0.118	0.624	0.251
T078.1	0.939	0.873	0.912	0.957	0.842	0.855	0.755
T079.1	0.732	0.728	0.656	0.956	0.645	0.956	0.206
T080.1	0.982	0.978	0.979	0.980	0.977	0.978	0.888
T081.1	0.343	0.319	0.289	0.613	0.246	0.599	0.645
T085.1	0.929	0.919	0.911	0.958	0.897	0.948	0.789
T086.1	0.797	0.790	0.711	0.936	0.688	0.936	0.742
T086.2	0.665	0.665	0.542	0.971	0.542	0.971	0.031
T087.1	0.971	0.963	0.972	0.963	0.963	0.956	0.903
T088.1	0.828	0.835	0.734	0.823	0.770	0.838	0.302
T089.1	0.924	0.895	0.883	0.976	0.840	0.958	0.903
T090.1	0.978	0.975	0.971	0.972	0.973	0.976	0.880
T091.1	0.952	0.940	0.948	0.953	0.932	0.939	0.911
T092.1	0.961	0.958	0.957	0.961	0.955	0.959	0.902
T093.1	0.974	0.969	0.933	0.972	0.926	0.981	0.892
T094.1	0.898	0.892	0.878	0.943	0.857	0.939	0.748
T095.1	0.656	0.337	0.678	0.700	0.338	0.305	0.586
T0960o	0.900	0.896	0.822	0.913	0.814	0.923	0.852
T096.1	0.663	0.650	0.618	0.688	0.600	0.677	-0.024
T0966o	0.769	0.768	0.324	0.835	0.324	0.834	-0.374
T097.1	0.858	0.859	0.851	0.848	0.854	0.850	0.243
T098.1	0.769	0.759	0.687	0.859	0.665	0.864	0.360
T099.1	0.789	0.747	0.752	0.761	0.742	0.636	0.611
T100.1	0.731	0.733	0.683	0.675	0.775	0.517	0.568
T1003o	0.650	0.612	0.658	0.660	0.608	0.610	0.668
T1009o	0.858	0.860	0.812	0.930	0.815	0.934	0.517
T101.1	0.516	0.503	0.315	0.727	0.286	0.704	0.128
T1016o	0.888	0.827	0.883	0.875	0.824	0.830	0.785
T1032o	0.947	0.875	0.946	0.964	0.864	0.893	0.910
T1038o	0.441	0.416	0.470	0.370	0.419	0.345	0.017
T104.1	0.968	0.973	0.960	0.955	0.972	0.973	0.904
T105.1	0.966	0.962	0.959	0.962	0.960	0.963	0.941
T1054o	0.533	0.517	0.298	0.737	0.268	0.721	0.164
T107.1	0.888	0.889	0.811	0.970	0.810	0.971	0.234
T1073o	0.490	0.497	0.366	0.660	0.366	0.669	0.051
T1080o	0.930	0.921	0.896	0.953	0.879	0.945	0.649
T110.1	0.958	0.956	0.953	0.966	0.953	0.976	0.918
T1109o	0.797	0.742	0.806	0.820	0.729	0.752	0.792
T1110o	0.850	0.887	0.825	0.833	0.874	0.899	0.776
T1113o	0.969	0.896	0.958	0.955	0.900	0.891	0.806
T1115o	-0.052	-0.052	-0.052	-0.440	-0.052	-0.440	-0.440
T1121o	0.578	0.600	0.577	0.505	0.662	0.558	0.208
T1123o	0.970	0.977	0.956	0.964	0.969	0.971	0.910

T1124o	0.904	0.889	0.902	0.882	0.894	0.876	0.737
T1127o	0.894	0.892	0.877	0.885	0.879	0.904	0.833
T113.1	0.807	0.805	0.585	0.842	0.559	0.838	0.422
T1132o	0.981	0.967	0.965	0.980	0.966	0.967	0.940
T1153o	0.879	0.949	0.816	0.793	0.947	0.953	0.696
T1160o	0.084	0.073	0.052	0.082	0.035	0.089	0.035
T1161o	0.512	0.550	0.486	0.489	0.547	0.541	0.342
T116.1	0.766	0.659	0.772	0.806	0.568	0.708	0.872
T1170o	0.745	0.426	0.743	0.730	0.416	0.420	0.682
T117.1	0.870	0.827	0.749	0.974	0.626	0.948	0.750
T117.2	0.694	0.685	0.470	0.838	0.452	0.822	0.021
T1173o	0.739	0.790	0.694	0.642	0.814	0.751	0.650
T117.3	0.874	0.826	0.745	0.967	0.649	0.935	0.791
T1174o	0.887	0.917	0.889	0.846	0.924	0.881	0.743
T117.4	0.567	0.567	0.461	0.950	0.461	0.950	0.216
T1176o	0.601	0.506	0.662	0.447	0.578	0.240	0.417
T1176v1o	0.557	0.524	0.544	0.642	0.495	0.640	0.250
T1178o	0.926	0.939	0.925	0.915	0.950	0.923	0.905
T1179o	0.968	0.944	0.971	0.961	0.946	0.936	0.877
T1181o	0.932	0.885	0.935	0.928	0.879	0.889	0.886
T1187o	0.948	0.947	0.895	0.967	0.892	0.966	0.625
T119.1	0.972	0.952	0.964	0.972	0.946	0.954	0.914
T1192o	0.424	0.230	0.468	0.349	0.381	-0.038	0.438
T1201o	0.916	0.934	0.882	0.897	0.938	0.930	0.768
T120.1	0.954	0.949	0.914	0.958	0.907	0.965	0.774
T1206o	0.874	0.871	0.773	0.793	0.870	0.872	0.631
T121.1	0.447	0.307	0.239	0.840	-0.041	0.840	0.799
T122.1	0.876	0.794	0.871	0.881	0.844	0.511	0.716
T123.1	0.291	0.274	0.194	0.385	0.171	0.368	0.217
T1234o	0.873	0.893	0.870	0.867	0.907	0.877	0.830
T1235o	0.827	0.821	0.868	0.788	0.920	0.748	0.815
T1237o	0.877	0.753	0.824	0.861	0.754	0.755	0.748
T1240o	0.821	0.853	0.786	0.737	0.851	0.842	0.347
T124.1	0.523	0.495	0.430	0.555	0.414	0.416	0.425
T125.1	0.943	0.931	0.938	0.948	0.918	0.937	0.840
T125.2	0.929	0.905	0.895	0.928	0.904	0.908	0.826
T125.3	0.789	0.784	0.709	0.816	0.702	0.810	-0.027
T125.4	0.264	0.266	0.259	0.268	0.261	0.247	0.190
T1298o	0.919	0.887	0.923	0.916	0.901	0.871	0.767
T131.1	0.795	0.789	0.711	0.960	0.691	0.958	0.303
T132.1	0.826	0.808	0.656	0.986	0.596	0.985	0.214
T133.1	0.962	0.959	0.941	0.981	0.927	0.981	0.829
T136.1	0.842	0.777	0.828	0.842	0.759	0.803	0.802
T136.2	0.922	0.801	0.927	0.940	0.793	0.815	0.888
T136.3	0.815	0.777	0.797	0.808	0.766	0.789	0.206
T138.1	0.676	0.654	0.526	0.917	0.500	0.917	0.213
T138.2	0.657	0.657	0.576	0.889	0.574	0.886	0.210
T140.1	0.962	0.961	0.955	0.971	0.953	0.972	0.948
T141.1	0.773	0.765	0.695	0.780	0.692	0.777	0.687
T142.1	0.825	0.814	0.787	0.957	0.777	0.956	0.699
T143.1	0.865	0.845	0.834	0.828	0.842	0.855	0.774
T144.1	0.944	0.874	0.959	0.962	0.869	0.878	0.867
T146.1	0.900	0.896	0.870	0.898	0.870	0.907	0.786
T146.2	0.812	0.816	0.646	0.949	0.631	0.958	0.264
T146.3	0.188	0.175	0.101	0.937	0.081	0.936	0.087
T149.1	0.902	0.897	0.888	0.982	0.880	0.978	0.817
T149.2	0.066	0.066	-0.033	0.325	-0.033	0.325	0.244
T149.3	0.879	0.805	0.906	0.917	0.803	0.815	0.897
T149.4	0.422	0.422	0.143	0.731	0.143	0.731	-0.056
T149.5	0.747	0.745	0.563	0.846	0.558	0.840	0.061
T150.1	0.912	0.908	0.879	0.987	0.875	0.982	0.667
T150.2	0.173	0.173	-0.001	0.428	-0.020	0.428	-0.032
T150.3	0.903	0.838	0.930	0.924	0.802	0.828	0.811
T150.4	0.403	0.402	0.241	0.689	0.232	0.687	-0.011
T150.5	0.897	0.904	0.581	0.910	0.522	0.902	0.022
T151.1	0.910	0.878	0.894	0.987	0.858	0.961	0.817
T151.2	0.074	0.074	-0.168	0.371	-0.185	0.370	0.257
T151.3	0.949	0.791	0.964	0.969	0.797	0.794	0.794
T151.4	0.397	0.397	0.255	0.718	0.255	0.718	0.251
T151.5	0.541	0.541	0.271	0.713	0.214	0.708	0.207

T152.1	0.805	0.772	0.806	0.822	0.763	0.785	0.824
T153.1	0.967	0.969	0.952	0.966	0.958	0.985	0.856
T154.1	0.875	0.840	0.807	0.942	0.798	0.944	0.562
T158.1	0.943	0.932	0.935	0.944	0.924	0.939	0.901
T159.1	0.954	0.928	0.955	0.956	0.924	0.932	0.961
T159.2	0.767	0.661	0.770	0.770	0.609	0.662	0.731
T159.3	0.816	0.754	0.683	0.959	0.610	0.940	0.393
T159.4	0.764	0.763	0.606	0.800	0.604	0.798	-0.273
T159.5	0.759	0.756	0.715	0.815	0.712	0.813	-0.223
T159.6	0.927	0.812	0.939	0.959	0.811	0.832	0.918
T159.7	0.931	0.864	0.910	0.949	0.814	0.878	0.653
T164.1	0.917	0.829	0.922	0.933	0.810	0.856	0.891
T165.1	0.082	0.052	0.065	0.339	-0.001	0.037	0.677
T166.1	0.938	0.929	0.923	0.938	0.917	0.937	0.885
T169.1	0.438	0.431	0.104	0.729	0.086	0.726	-0.105
T176.1	0.635	0.583	0.469	0.840	0.352	0.843	0.386
T180.1	0.810	0.771	0.808	0.825	0.762	0.776	0.692
T180.2	0.921	0.830	0.932	0.949	0.811	0.851	0.922
Average	0.744	0.721	0.692	0.792	0.664	0.757	0.533

Table S8. Average running time (RT) in seconds with standard deviation (SD) achieved by TM-score-Comp, OpenStructure, and DockQ on each CASP/CAPRI target

Target	TM-score-Comp		OpenStructure		DockQ	
	RT	SD	RT	SD	RT	SD
H0953	0.082	0.018	4.112	1.100	0.448	0.067
H0957	0.046	0.007	3.000	0.812	0.125	0.028
H0968	0.067	0.034	4.284	1.288	0.371	0.092
H1036	1.740	1.024	47.177	18.607	6.746	9.871
H1072	0.055	0.018	3.104	0.889	0.348	0.063
H1106	0.015	0.002	2.649	8.240	0.072	0.027
H1111	2.715	3.323	400.509	455.836	4.158	0.648
H1114	4.067	9.383	454.690	894.478	450.806	2637.635
H1114v2	0.268	0.114	18.032	14.908	2.627	4.080
H1129	0.678	0.267	21.546	10.686	0.449	0.052
H1134	0.110	0.028	5.538	9.602	0.187	0.035
H1135	1.176	0.614	39.785	20.686	752.402	1705.935
H1137	3.664	1.803	150.657	79.486	3.818	0.919
H1140	0.050	0.008	3.850	9.296	0.128	0.026
H1141	0.044	0.008	3.369	5.691	0.128	0.021
H1142	0.047	0.007	3.337	5.868	0.128	0.022
H1143	0.047	0.016	3.728	5.903	0.129	0.019
H1144	0.040	0.010	3.578	6.241	0.119	0.016
H1151	0.017	0.002	2.004	3.126	0.129	1.011
H1157	0.832	0.251	22.713	8.255	0.491	0.058
H1166	0.138	0.049	5.816	7.144	0.246	0.877
H1166v1	0.237	0.068	5.811	7.925	0.225	0.112
H1167	0.124	0.047	5.846	8.894	0.206	0.077
H1167v1	0.266	0.079	5.813	9.672	0.217	0.102
H1168	0.102	0.028	6.123	8.980	0.221	0.075
H1168v1	0.170	0.027	5.882	9.223	0.236	0.097
H1185	0.243	0.066	12.026	6.609	0.467	0.044
H1202	0.044	0.008	3.979	9.676	0.361	0.045
H1204	0.259	0.074	9.566	9.413	0.936	0.128
H1213	0.372	0.103	15.637	9.806	1.580	0.208
H1222	0.108	0.024	6.362	8.948	0.184	0.030
H1223	0.137	0.061	6.802	10.317	0.188	0.027
H1225	0.106	0.040	6.323	9.047	0.184	0.025
H1232	0.223	0.073	9.649	9.249	0.939	0.115
H1233	0.246	0.057	10.062	9.187	1.141	0.148
H1236	0.957	0.247	39.727	14.060	100.018	14.758
H1272	2.159	0.691	130.432	73.423	1.824	0.371
T024.1	0.025	0.004	2.121	0.807	0.119	0.023
T025.1	0.029	0.006	2.436	0.795	0.129	0.028
T026.1	0.088	0.017	4.395	1.009	0.179	0.043
T029.1	0.085	0.017	4.455	1.097	0.877	13.262
T030.1	0.023	0.003	2.171	0.710	0.124	0.061
T032.1	0.061	0.014	4.315	1.011	0.149	0.044
T035.1	0.084	0.034	4.691	1.234	0.236	0.107
T036.1	0.075	0.029	4.300	1.146	0.248	0.106

T037.1	0.025	0.008	2.278	0.799	0.343	0.050
T037.2	0.025	0.009	2.306	0.780	0.341	0.045
T038.1	0.066	0.013	4.389	1.067	0.171	0.064
T039.1	0.063	0.012	4.576	1.050	0.719	10.735
T040.1	0.043	0.011	3.898	0.979	0.141	0.046
T040.2	0.043	0.014	3.861	1.069	0.144	0.048
T041.1	0.017	0.005	1.860	0.776	0.168	0.071
T042.1	0.018	0.002	1.494	0.400	0.228	0.051
T046.1	0.037	0.005	2.377	0.733	0.330	4.286
T047.1	0.017	0.003	1.874	0.688	0.228	0.044
T050.1	0.084	0.026	4.101	1.111	0.628	0.066
T053.1	0.028	0.005	2.188	0.677	0.121	0.072
T054.1	0.015	0.003	1.660	0.691	0.072	0.049
T058.1	0.030	0.008	2.578	0.697	0.128	0.051
T060.1	0.051	0.018	4.011	0.927	0.378	0.085
T060.2	0.054	0.017	4.288	0.934	0.157	0.057
T060.3	0.051	0.017	4.045	0.922	0.380	0.080
T061.1	0.050	0.015	4.004	0.861	0.389	0.088
T061.2	0.053	0.016	4.143	0.962	0.390	0.139
T061.3	0.051	0.015	4.087	0.918	0.397	0.099
T062.1	0.050	0.015	4.244	0.966	0.398	0.132
T062.2	0.052	0.015	4.229	0.891	0.391	0.122
T062.3	0.050	0.015	4.142	0.858	0.408	0.133
T063.1	0.052	0.019	4.081	0.879	0.390	0.091
T063.2	0.055	0.019	4.306	0.905	0.167	0.116
T063.3	0.053	0.019	4.154	0.917	0.400	0.135
T064.1	0.051	0.015	4.065	0.918	0.362	0.162
T064.2	0.054	0.015	4.304	0.997	0.163	0.055
T064.3	0.042	0.011	3.888	0.926	0.344	0.110
T065.1	0.011	0.001	1.418	0.720	0.250	0.044
T065.2	0.011	0.001	1.442	0.685	0.079	0.062
T066.1	0.093	0.030	3.332	0.949	0.424	0.053
T066.2	0.112	0.029	4.684	1.315	0.439	0.062
T066.3	0.123	0.042	4.817	1.457	0.451	0.078
T067.1	0.001	0.000	0.793	0.653	0.038	0.101
T067.2	0.001	0.000	0.762	0.646	0.032	0.096
T068.1	0.019	0.003	1.709	0.655	0.258	0.032
T068.2	0.018	0.003	1.723	0.682	0.238	0.040
T069.1	0.155	0.036	5.654	1.107	0.585	0.173
T070.1	0.013	0.002	1.429	0.594	0.254	0.143
T070.2	0.015	0.003	1.446	0.606	0.259	0.166
T071.1	0.033	0.004	2.443	0.691	0.291	0.135
T071.2	0.031	0.003	2.587	0.715	0.287	0.066
T071.3	0.033	0.004	2.646	0.715	0.293	0.146
T072.1	0.327	0.062	8.374	2.002	0.994	2.352
T073.1	0.096	0.022	3.620	0.817	0.386	0.077
T073.2	0.099	0.027	3.941	0.781	0.381	0.074
T075.1	0.081	0.018	4.528	1.046	0.675	3.512
T077.1	0.072	0.015	3.559	0.989	0.359	0.058

T078.1	0.096	0.020	5.367	1.287	0.472	0.189
T079.1	0.014	0.002	1.339	0.578	0.236	0.129
T080.1	0.178	0.047	6.578	1.476	0.722	0.177
T081.1	0.021	0.002	2.349	0.710	0.283	0.069
T085.1	0.128	0.031	5.034	1.190	0.531	0.094
T086.1	0.020	0.002	1.854	0.690	0.259	0.120
T086.2	0.019	0.002	1.785	0.695	0.263	0.120
T087.1	0.192	0.037	6.694	1.492	0.593	0.145
T088.1	0.050	0.014	3.474	0.771	0.336	0.122
T089.1	0.325	0.101	8.788	1.703	0.278	0.052
T090.1	0.265	0.080	7.542	1.485	0.767	0.136
T091.1	0.054	0.010	3.501	0.931	0.327	0.140
T092.1	0.101	0.018	5.020	0.946	0.458	0.140
T093.1	0.423	0.178	9.615	1.918	0.910	0.123
T094.1	0.263	0.089	7.799	1.917	0.747	0.180
T095.1	0.267	0.081	10.336	2.277	3.897	0.421
T0960o	0.568	0.181	11.735	4.206	1.073	0.136
T096.1	0.062	0.018	4.884	1.259	0.164	0.034
T0966o	0.412	0.123	9.330	1.855	0.686	0.099
T097.1	0.039	0.009	3.611	0.906	0.184	0.077
T098.1	0.047	0.007	3.652	0.943	0.197	0.083
T099.1	0.063	0.009	5.069	0.990	0.273	0.131
T100.1	0.054	0.009	4.170	0.978	0.207	0.137
T1003o	0.284	0.131	9.797	2.993	0.654	0.081
T1009o	0.849	0.335	22.289	8.585	1.034	0.116
T101.1	0.080	0.016	3.814	0.787	0.236	0.486
T1016o	0.108	0.024	4.352	0.998	0.348	0.048
T1032o	0.100	0.027	4.727	1.179	0.346	0.043
T1038o	0.068	0.012	3.481	0.922	0.295	0.045
T104.1	0.018	0.004	1.904	0.635	0.222	0.038
T105.1	0.017	0.003	1.817	0.557	0.075	0.097
T1054o	0.045	0.013	2.776	0.863	0.252	0.030
T107.1	0.228	0.031	9.128	1.594	0.400	0.233
T1073o	0.034	0.004	3.880	1.226	0.538	0.065
T1080o	0.099	0.016	10.711	6.544	0.759	0.355
T110.1	0.033	0.004	2.369	0.613	0.252	0.132
T1109o	0.101	0.015	5.433	9.016	0.373	0.053
T1110o	0.092	0.013	5.367	8.345	0.406	0.052
T1113o	0.081	0.017	3.719	0.878	0.313	0.041
T1115o	3.815	3.737	226.398	158.970	1263.670	2274.959
T1121o	0.326	0.082	7.021	1.686	0.688	0.099
T1123o	0.127	0.031	5.709	6.243	0.422	0.061
T1124o	0.227	0.061	8.193	5.955	0.684	0.111
T1127o	0.064	0.006	4.997	8.095	0.381	0.041
T113.1	0.014	0.002	1.468	0.501	0.067	0.113
T1132o	0.135	0.038	6.257	1.266	8.807	1.277
T1153o	0.125	0.030	5.106	1.176	0.512	0.560
T1160o	0.004	0.000	0.543	0.313	0.107	0.022
T1161o	0.008	0.001	0.749	0.402	0.140	0.022

T116.1	0.091	0.013	4.830	1.021	0.490	1.618
T1170o	1.777	0.578	42.573	10.036	9.537	1.959
T117.1	0.111	0.019	5.447	0.969	0.216	0.064
T117.2	0.133	0.016	4.555	0.930	0.514	0.226
T1173o	0.174	0.046	7.222	9.885	0.579	0.085
T117.3	0.118	0.021	5.471	1.086	0.204	0.037
T1174o	0.682	0.173	13.397	8.600	1.004	0.138
T117.4	0.150	0.042	5.816	1.240	0.587	0.110
T1176o	0.407	0.292	17.698	7.198	558.569	125.492
T1176v1o	0.045	0.009	5.340	9.695	0.401	0.301
T1178o	0.134	0.029	5.860	5.743	0.477	0.060
T1179o	0.171	0.046	5.240	2.511	0.445	0.054
T1181o	1.621	0.489	43.578	13.254	2.095	0.370
T1187o	0.059	0.010	3.366	5.451	0.254	0.039
T119.1	0.251	0.070	8.207	1.935	0.745	0.670
T1192o	1.044	0.397	29.748	11.012	24405.129	11258.258
T1201o	0.045	0.003	3.318	3.451	0.303	0.043
T120.1	0.020	0.003	1.974	0.676	0.072	0.049
T1206o	0.062	0.023	4.864	5.964	0.363	0.054
T121.1	0.007	0.001	1.173	0.613	0.049	0.036
T122.1	0.158	0.038	6.148	1.509	0.250	0.034
T123.1	0.024	0.004	1.519	0.316	0.113	0.033
T1234o	0.521	0.149	15.862	8.692	1.208	0.201
T1235o	0.226	0.072	7.189	6.202	7.044	1.054
T1237o	1.134	0.515	37.205	13.005	3.462	0.558
T1240o	0.392	0.145	12.441	8.222	1.000	0.188
T124.1	0.101	0.017	3.677	0.879	0.369	0.078
T125.1	0.019	0.004	2.139	0.754	0.104	0.043
T125.2	0.015	0.005	1.810	0.705	0.229	0.054
T125.3	0.018	0.002	2.099	0.702	0.109	0.049
T125.4	0.021	0.007	1.657	0.695	0.236	0.103
T1298o	0.192	0.028	6.928	8.339	0.497	0.090
T131.1	0.045	0.006	3.873	0.909	0.144	0.041
T132.1	0.054	0.008	3.910	0.892	0.164	0.058
T133.1	0.016	0.003	1.877	0.678	0.074	0.039
T136.1	0.728	0.155	23.309	5.567	1.542	0.116
T136.2	0.729	0.142	23.265	5.196	1.551	0.117
T136.3	0.719	0.126	22.652	5.747	1.492	0.113
T138.1	0.366	0.053	10.353	1.739	0.932	0.194
T138.2	0.367	0.058	9.297	1.534	0.883	0.175
T140.1	0.033	0.007	2.496	0.699	0.271	0.069
T141.1	0.105	0.031	5.100	1.493	0.491	0.146
T142.1	0.009	0.002	1.401	0.615	0.053	0.023
T143.1	0.073	0.011	5.022	0.935	0.442	0.102
T144.1	0.915	0.268	20.308	4.624	1.023	0.272
T146.1	0.118	0.041	5.327	1.446	0.469	0.094
T146.2	0.044	0.009	3.403	0.848	0.134	0.043
T146.3	0.043	0.008	3.234	0.775	0.135	0.043
T149.1	0.226	0.056	12.357	3.905	1.476	0.552

T149.2	0.478	0.314	16.335	8.007	1.570	0.478
T149.3	0.241	0.050	8.447	2.523	0.270	0.055
T149.4	0.233	0.039	8.035	1.406	0.298	0.060
T149.5	0.191	0.050	6.065	1.423	0.283	0.124
T150.1	0.222	0.059	12.928	3.362	1.538	0.501
T150.2	0.511	0.301	17.386	7.034	1.610	0.446
T150.3	0.255	0.063	9.507	2.717	0.272	0.050
T150.4	0.234	0.037	8.716	1.187	0.294	0.055
T150.5	0.166	0.031	6.215	1.437	0.251	0.112
T151.1	0.203	0.039	12.569	3.495	1.493	0.435
T151.2	0.419	0.239	17.498	7.546	1.620	0.512
T151.3	0.250	0.054	9.626	2.554	0.271	0.063
T151.4	0.222	0.054	8.244	1.226	0.305	0.075
T151.5	0.153	0.048	5.731	1.098	0.296	0.135
T152.1	0.242	0.106	8.269	1.894	0.811	0.168
T153.1	0.010	0.002	1.314	0.596	0.211	0.064
T154.1	0.767	0.256	19.437	6.484	1.337	0.284
T158.1	0.233	0.072	6.727	1.381	0.733	0.175
T159.1	0.040	0.007	2.870	0.677	0.277	0.060
T159.2	0.116	0.037	4.553	1.057	0.472	0.094
T159.3	0.119	0.008	5.100	0.871	0.213	0.052
T159.4	0.065	0.010	4.170	0.537	0.150	0.053
T159.5	0.064	0.007	4.219	0.569	0.145	0.051
T159.6	0.239	0.040	7.126	1.471	0.587	0.117
T159.7	0.111	0.019	6.182	1.158	0.171	0.054
T164.1	0.097	0.023	4.476	1.092	0.404	0.081
T165.1	0.242	0.087	7.628	1.421	0.296	0.081
T166.1	0.045	0.012	3.223	0.850	0.122	0.024
T169.1	0.046	0.014	2.633	0.779	0.275	0.059
T176.1	0.036	0.008	2.390	0.815	0.250	0.056
T180.1	0.092	0.022	4.370	1.055	0.415	0.092
T180.2	0.094	0.024	4.504	1.009	0.382	0.080
Average	0.252	0.141	12.894	10.503	123.537	80.793

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